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Scaling Community Storytelling with Open-Source AI

Building an End-to-End Video Processing Pipeline on Red Hat OpenShift AI

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Storykeeping Collective preserves African American oral histories through documentary storytelling.

They record interviews with elders, veterans, community leaders, and families to capture stories often missing from traditional historical records.

The Challenge

- Hundreds of interviews have been recorded, but turning each one into a documentary is a slow, manual process.
- Many valuable stories remain archived and unseen.

What I Built

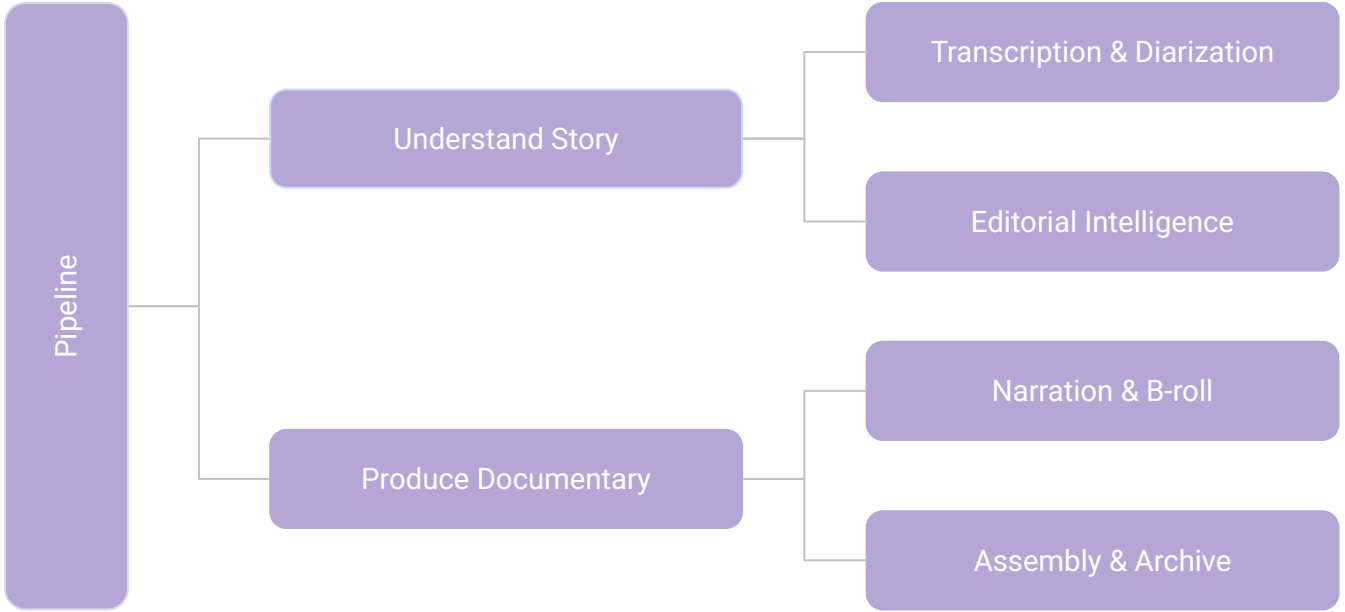
An automated pipeline that handles the production workflow from beginning to end.

- **End-to-end automation**
  Raw interview in → completed documentary out
- **Story-first output**
  Preserves the subject's own voice while adding narration and supporting visuals
- **Built to scale**
  Designed to process a growing archive without requiring manual editing for every interview

Raw Interview → Automated Production → Mini-Documentary

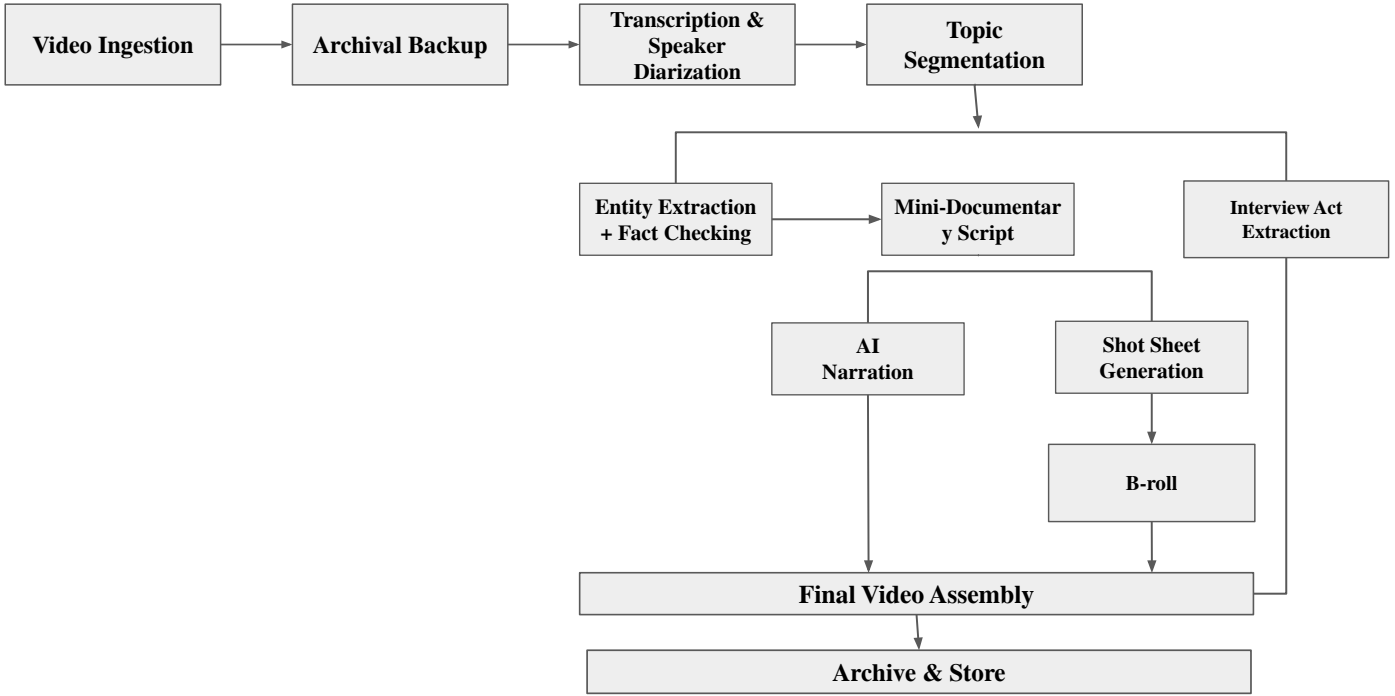
Pipeline Architecture

11-Stage End-to-End Workflow



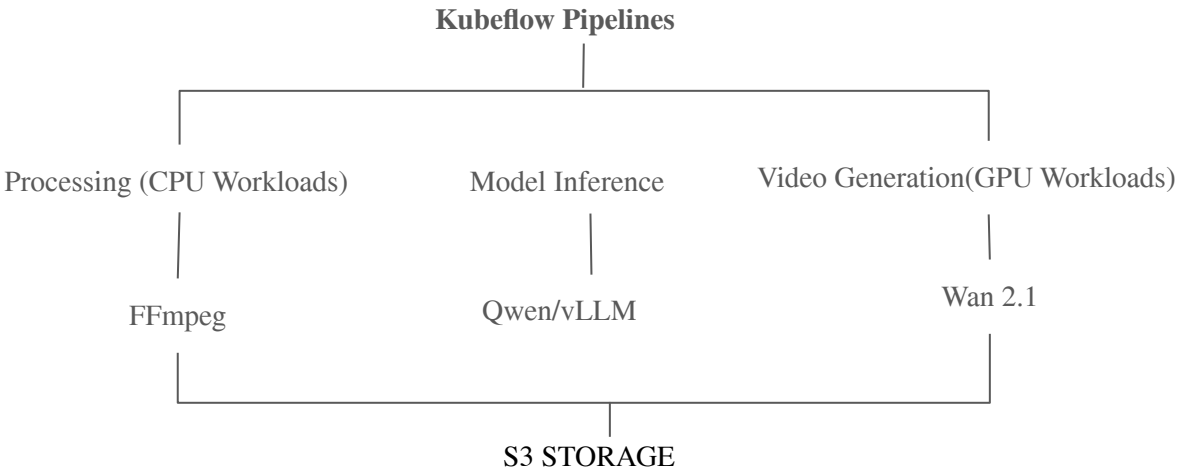
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From Interview to Documentary



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Running the Pipeline on OpenShift AI



Open-Source Stack

Different models for different parts of production

WhisperX + pyannote	Transcription & speaker diarization
Qwen 2.5 7B + vLLM	Editorial and reasoning tasks
Chatterbox	AI narration
Wan 2.1	Generative B-roll
FFmpeg	Clip extraction & final assembly

Demo



Engineering Challenges & Fixes

Things I Learned While Building It

Challenge	Fix
Interviewer speech leaked into clips	Speaker-aware sub-segment extraction
Invalid LLM JSON responses	Preprocessing + response-repair layer
Inconsistent B-roll realism	Photorealistic + historical prompting
Slow video generation	Parallel generation across GPU pods

Thank you!

